

CPU Microbenchmark Suite

Measuring What Spec Sheets Only Claim

Cache Latency, STREAM Bandwidth, and Peak SIMD FLOPS
on an Intel Core i7-14700K

Olajide Badejo

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Abstract

This report measures, on real hardware, the numbers usually quoted from specification sheets: cache load use latencies, sustainable memory bandwidth, and peak single precision floating point throughput. Every figure is measured on an Intel Core i7-14700K running inside WSL2, using timing and known transfer sizes rather than hardware performance counters, which WSL2 does not expose. A prefetcher defeating pointer chase recovers the L1, L2, L3, and DRAM plateaus and cross checks them against sysfs. STREAM in three variants gives a bandwidth scaling curve that saturates well before the cores do. A ten chain FMA loop reaches the peak throughput computed from a software measured clock, to within one percent. The ceilings are assembled into a measured roofline, and a BLIS style analytical tile prediction is validated against a blocked GEMM sweep. The prediction misses the naive kernel by design, and that miss is reported as a finding rather than hidden.

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Chapter 1

Introduction

Specification sheets quote cache latencies in cycles, memory bandwidth in gigabytes per second, and peak floating point throughput in GFLOPS. These numbers are ceilings under idealized assumptions. What a program actually sees depends on the access pattern, the compiler, the memory hierarchy under load, and the clock the core really runs at. This project measures those numbers directly on one machine, an Intel Core i7-14700K, and assembles them into a model that predicts something concrete: the cache block sizes a blocked matrix multiply should use.

The work has five measured pillars. First, a dependent load pointer chase recovers the load use latency of each cache level, from a 4 KB working set that lives in L1 up to a 512 MB working set that is firmly in DRAM, and a change point detector locates the plateau boundaries and cross checks them against the cache sizes the operating system reports. Second, a STREAM style benchmark in three variants measures sustainable bandwidth and a thread scaling curve. Third, a ten chain fused multiply add loop measures peak single precision throughput and compares it to the ceiling computed from the measured clock. Fourth, these ceilings become a roofline, and a BLIS style analytical model predicts matrix multiply tile sizes that a blocked GEMM sweep then tests. Fifth, the compute kernels are ported to AArch64 NEON and checked for correctness under emulation.

The guiding principle is measurement honesty. The target runs inside WSL2, which exposes no hardware performance counters, so every number here comes from timing and a known transfer size, the classical approach of McCalpin [1]. Where the platform makes a number hard to get right, this report states the difficulty and the workaround rather than quietly publishing a plausible figure. Two such difficulties, described in the methodology, turned out to matter a great deal: the reported clock does not track the boost frequency, and the hybrid core layout is invisible to the guest.

Chapter 2

Background

2.1 The memory hierarchy and load use latency

A modern core hides memory latency with out of order execution: while one load is outstanding, independent work continues. This makes ordinary array traversal a poor latency probe, because the hardware overlaps many accesses. The honest probe is a dependent load chain, where the address of the next load is the value returned by the previous one. The processor cannot start load $n + 1$ until load n retires, so the measured time per load is the true load use latency of whichever cache level currently holds the data. Drepper [2] and Hennessy and Patterson [3] describe this technique and the plateau structure it reveals.

Two things must be defeated for the probe to be honest. The stride prefetcher will predict a regular access pattern and hide the latency, so the chase visits a random permutation at a 64 byte (one cache line) stride, which the prefetcher cannot anticipate. The permutation must also form a single cycle over all nodes; a permutation that splits into shorter cycles would keep a small footprint resident in L1 regardless of the allocation size, silently reporting L1 latency everywhere.

2.2 The STREAM model and machine balance

McCalpin’s STREAM [1] measures sustainable bandwidth with four kernels, Copy, Scale, Add, and Triad, on arrays sized several times larger than the last level cache so the data cannot hide in cache. Bandwidth is bytes moved divided by time, where bytes moved uses a fixed convention per kernel. Non temporal stores matter here: an ordinary store first reads the target line for ownership, doubling write traffic, whereas a streaming store writes straight to memory, so the counted bytes match the real DRAM traffic.

2.3 The roofline model

The roofline [4] bounds attainable performance by the minimum of two ceilings: peak compute, and arithmetic intensity multiplied by peak bandwidth. Arithmetic intensity is FLOPs performed per byte of memory traffic. A kernel with low intensity is memory bound and sits on the sloped bandwidth line; a kernel with high intensity is compute bound and sits under the flat compute ceiling. The two lines meet at the ridge point.

2.4 BLIS blocking

The BLIS framework [5] structures matrix multiply as packed panels that are blocked to reside at specific cache levels. The analytical model [6] sizes the cache blocks, M_c , K_c , and N_c , from

the cache capacities and associativities so that a panel of A stays in L2 and a panel of B stays in L3 while a micro panel of B stays in L1. This report computes such a prediction from the measured cache geometry and tests it against an empirical tile sweep.

Chapter 3

Methodology

3.1 What timing derived numbers can and cannot claim

The target runs inside WSL2, a Hyper-V utility virtual machine, which exposes no hardware performance monitoring unit. Every number in this report is therefore derived from wall clock timing and a known transfer size or operation count. This is the classical STREAM methodology and it is entirely sufficient for latency, bandwidth, and throughput. What it cannot provide is a direct cycle or cache miss count from the hardware; a `perf_event` wrapper is compiled but is gated to bare metal Linux, where `is_available` returns true, and it stays dormant under WSL2.

3.2 The clock does not track turbo

Cycle denominated figures need the frequency the core actually ran at. Under WSL2 the `cpu MHz` field in `/proc/cpuinfo` is pinned to a base value near 3.42 GHz and never moves, even under full load, and no `cpufreq` interface exists. The pinned core, however, boosts to about 5.3 GHz. Using the base value would understate every cycle count by roughly a factor of 1.6.

The fix is to measure the running clock in software. A dependent chain of integer additions executes at one cycle per iteration, because each addition depends on the previous result and the loop counter runs on a separate port. Timing a known iteration count gives the frequency directly. Because the chain cannot run faster than one cycle per iteration, any perturbation only lengthens the time, so the maximum frequency over several warmed passes is the cleanest estimate. This method reports about 5.3 GHz under load, and with it the L1 latency lands at exactly the expected 5 cycles.

3.3 Turbo stability and the power plan

For a stable clock the Windows power plan maximum processor state is set to 99 percent, which caps turbo so the boost frequency does not wander. The software clock measurement is taken during every run regardless, so a session without the cap is still honest, only noisier.

3.4 Pinning and the hybrid layout

Benchmarks pin to a single logical CPU with `sched_setaffinity`, a P core by default. The i7-14700K is a hybrid part with 8 performance cores and 12 efficiency cores, but WSL2 does not expose the `cpu_core` and `cpu_atom` split, and it reports a uniform L2 size across all cores, so P and E cannot be recovered from sysfs. The topology reader detects the split when present

and otherwise falls back to the documented Intel enumeration, marking the result as heuristic rather than measured.

3.5 Reproducibility

Every run records the compiler and flags, kernel version, WSL flag, measured clock, pinned core, and git commit into the summary, so any figure can be traced to the conditions that produced it. The reproducibility block for the run behind this report is in Table 3.1.

Table 3.1: Reproducibility record for the committed run.

Compiler	g++ (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Kernel	6.18.33.2-microsoft-standard-WSL2
WSL2	True
Measured clock	5.34 GHz
Git commit	e5e4ed1

Chapter 4

Implementation

4.1 Anti optimization

The single greatest risk in a microbenchmark is the compiler deleting the work being timed. Link time optimization is forbidden for every benchmark target and the build system asserts it off, because the linker would otherwise prove a measured loop has no observable effect and remove it. Inside each loop a `do_not_optimize` barrier, implemented as inline assembly with a memory clobber, forces the result to be treated as observed. The pointer chase walking pointer, the STREAM outputs, and the FMA accumulators all pass through such barriers.

4.2 Pointer chase

Nodes are 64 byte aligned and linked by a Fisher Yates permutation whose visit order forms one Hamiltonian cycle by construction. A unit test proves the single cycle property for every swept size and rejects deliberately broken multi cycle inputs, so a green test means the guarantee has teeth. The working set sweeps from 4 KB to 512 MB in powers of two, and per point the timed window is sized adaptively to a fixed wall time so small and large working sets are both stable.

4.3 STREAM

The four kernels appear in three variants. The scalar baseline suppresses vectorization on those functions only, using a function attribute, so the honest scalar number is genuinely scalar while the rest of the translation unit stays optimized. The compiler vectorized variant is the same source at `-O3 -march=native`. The intrinsic variant uses AVX2 with `_mm256_stream_ps` non temporal stores for the DRAM test. Threads pin to P cores first and then E cores, with first touch initialization so pages are resident locally, and each worker times its own chunk between barriers to avoid a main thread timing race.

4.4 Peak FLOPS

The kernel runs ten independent FMA accumulator chains. Ten is chosen because the FMA units have a latency of four to five cycles across two pipes, so at least eight to ten in flight chains are needed to keep both pipes issuing every cycle. The multiplier is non unity, to prevent the compiler strength reducing the recurrence into an addition, and slightly less than one, so the values converge to a finite fixed point rather than drifting to infinity.

4.5 Blocked GEMM

The GEMM is a naive blocked triple loop with the innermost loop contiguous over N , so the compiler vectorizes it, blocked by (M_c, K_c, N_c) . It is deliberately not a packed micro kernel; the point is to see which tile size runs fastest, not to beat a tuned library. A correctness test checks the blocked result against a plain triple loop for several tiles, including tiles that do not divide the matrix dimensions.

4.6 AArch64 NEON port

The FLOPS and STREAM kernels have NEON paths behind `#ifdef __ARM_NEON`. The suite cross compiles with a static AArch64 toolchain and runs the correctness tests under `qemu-user`; timing is never taken under emulation. The FLOPS correctness test is written to be SIMD width independent, so one assertion covers both the 8 lane AVX2 build and the 4 lane NEON build.

Chapter 5

Results

All numbers below come from a single committed run on the i7-14700K, pinned to a P core, at a software measured clock of about 5.3 GHz.

5.1 Cache latency

The pointer chase recovers a clean four plateau staircase (Figure 5.1). L1 data latency is about 0.94 ns, which at the measured clock is 5.0 cycles, exactly the Raptor Cove L1 load use latency. L2 is about 16 cycles, the L3 region climbs through the tens of cycles, and DRAM settles near 93 ns. Table 5.1 lists the detected plateaus.

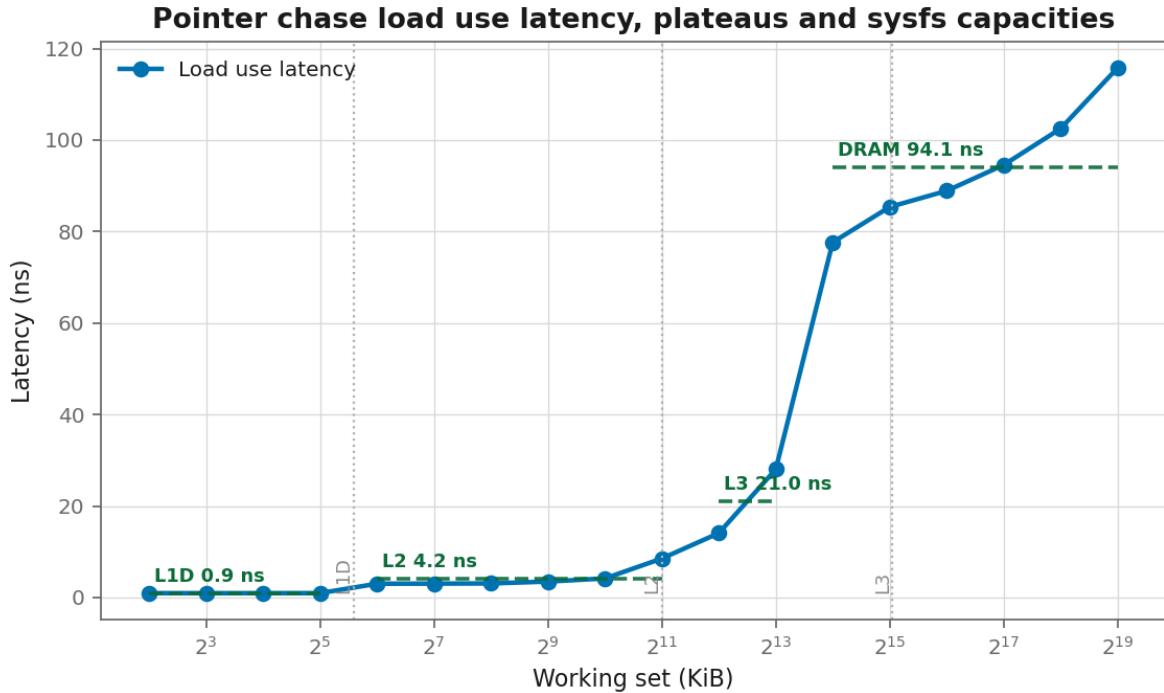


Figure 5.1: Load use latency against working set, with detected plateau means and the sysfs cache capacities marked. The step to DRAM begins about one octave below the nominal 33 MB L3, an effect of TLB reach and associativity under random access.

The change point detector places the L1 boundary within 0.08 octaves of the 48 KB L1 capacity and the L2 boundary within 0.50 octaves of the 2 MB L2. The DRAM onset is flagged about 1.5 octaves below the nominal L3 capacity, which the discussion addresses.

Table 5.1: Detected latency plateaus, cross checked against sysfs.

Region	Working set	Latency (ns)	Cycles
L1D	4 to 32 KiB	0.94	5.0
L2	64 to 2048 KiB	4.21	22.5
L3	4096 to 8192 KiB	21.04	112.3
DRAM	16384 to 524288 KiB	94.11	502.6

5.2 Bandwidth

Single thread bandwidth by variant is in Table 5.2 and Figure 5.2. The non temporal variant wins on Add and Triad, where avoiding the read for ownership pays off, while the scalar and vectorized variants are competitive on Copy. The thread scaling curve (Figure 5.3) saturates near 78 GB s^{-1} by four to five threads, about 87 percent of the DDR5-5600 dual channel theoretical 89.6 GB s^{-1} . Beyond that point additional P cores, and then the E cores, add no bandwidth: the reported knee is the bandwidth wall, not a P to E transition.

Table 5.2: Single thread STREAM bandwidth by variant, GB/s.

Variant	Copy	Scale	Add	Triad
Scalar	45.5	23.7	29.3	28.5
Auto vectorized	43.9	25.7	32.3	32.9
AVX2 NT stores	37.3	37.2	41.0	41.5

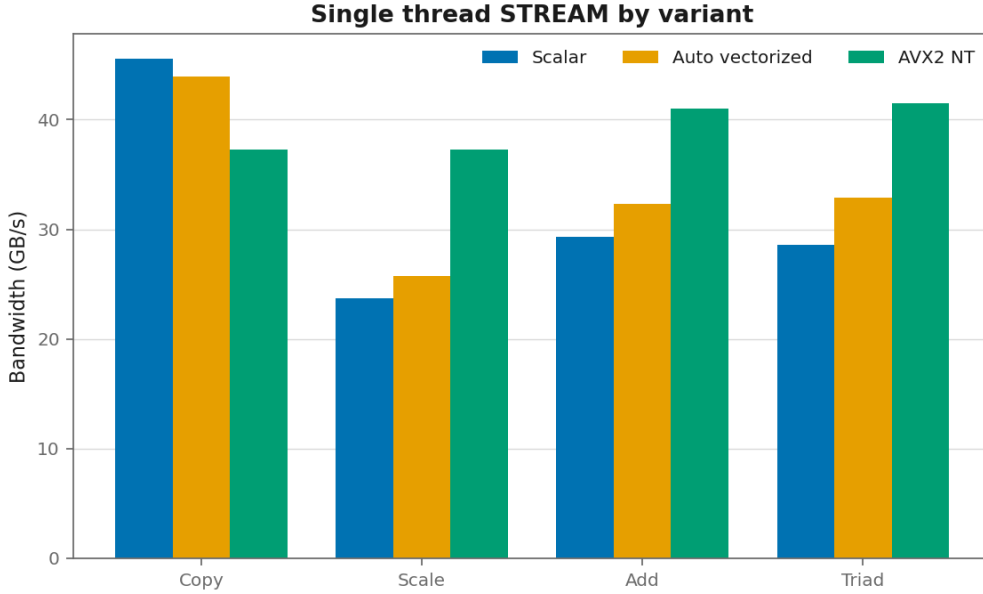


Figure 5.2: Single thread bandwidth by kernel and variant.

5.3 Peak FLOPS

Table 5.3 compares measured throughput to the ceiling computed from the same measured clock. One P core reaches about 170 GFLOPS, which is 99 percent of its ceiling; matching the ceiling to within one percent is the expected result for a saturating ten chain loop and

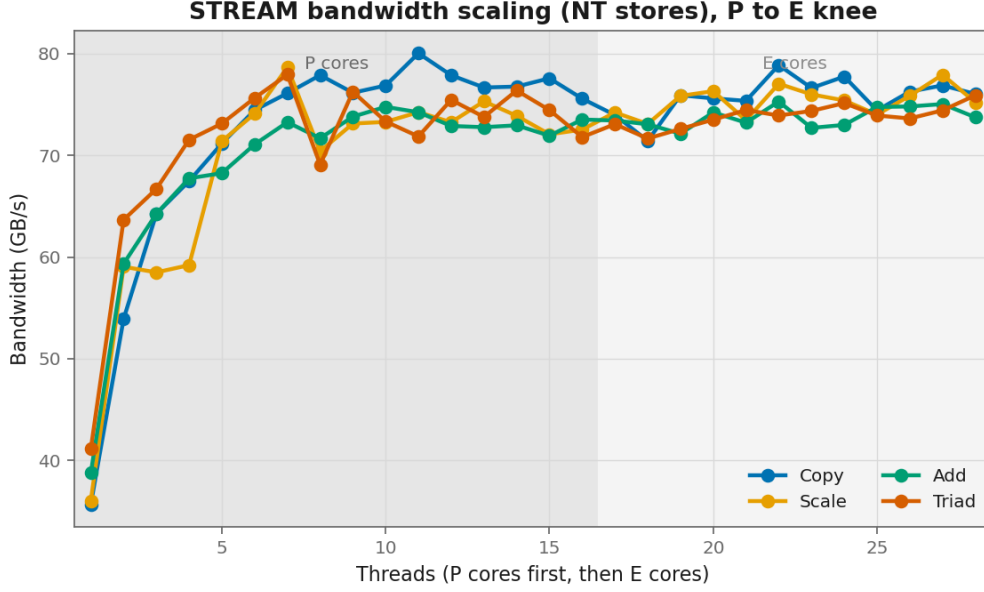


Figure 5.3: Bandwidth scaling, P cores first then E cores. The curve saturates well before the cores are exhausted.

confirms both the clock and the FLOP counting. All 28 logical CPUs together reach about 1900 GFLOPS.

Table 5.3: Peak single precision throughput against the computed ceiling.

Configuration	GFLOPS	Ceiling	Efficiency
Single P core	169.7	172.4	98%
All core (28 threads)	1878	(aggregate)	

5.4 Roofline and tile prediction

Figure 5.4 places the suite’s kernels on the measured roofline. STREAM Triad sits on the memory bound slope. The single core FMA loop sits on the single core ceiling. The naive blocked GEMM, though high in arithmetic intensity and therefore compute bound in principle, sits well below the single core ceiling: the gap is the cost of not packing.

Table 5.4 compares the BLIS analytical tile prediction to the empirical best from the GEMM sweep. They disagree: the model predicts a large K_c , the naive kernel prefers a much smaller one. This miss is a finding, not a failure, and the discussion explains it.

Table 5.4: BLIS analytical prediction versus empirical best tile.

Source	Mc	Kc	Nc
BLIS analytical	704	696	11296
Empirical best	384	64	1536

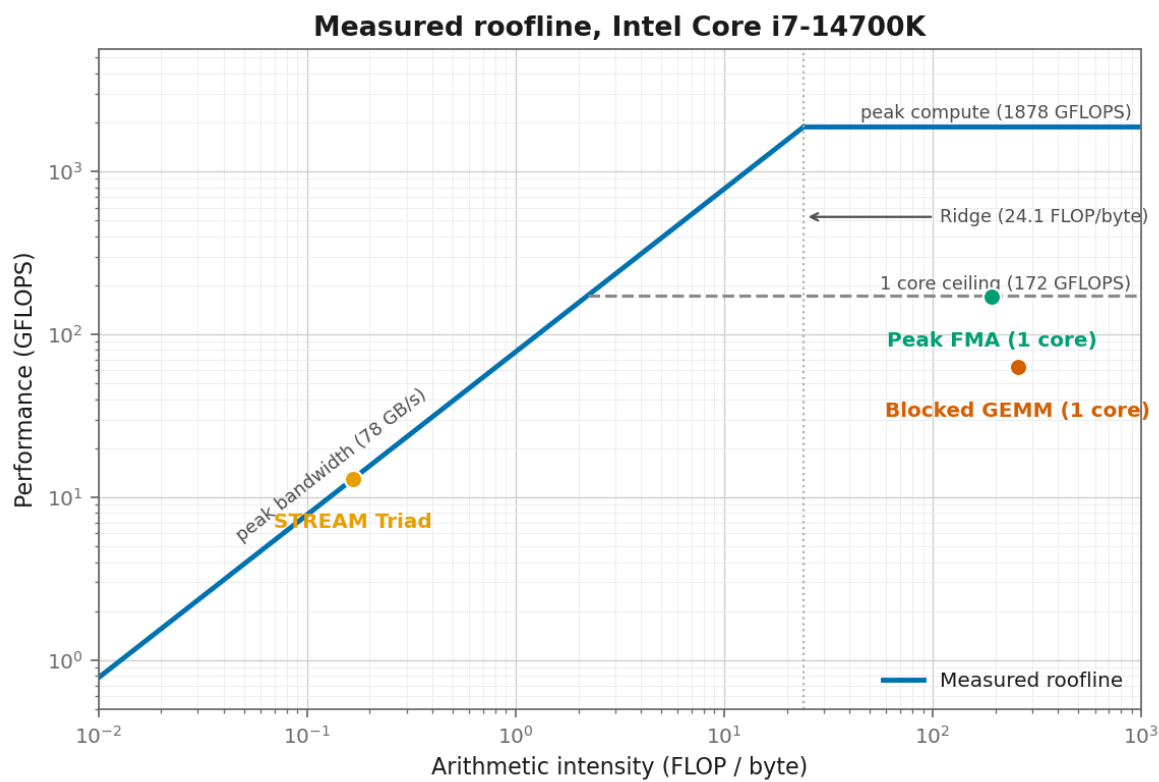


Figure 5.4: Measured roofline with all core and single core compute ceilings.

Chapter 6

Discussion

6.1 The DRAM onset below nominal L3

The latency staircase steps up to DRAM near an 8 to 16 MB working set, while the L3 is 33 MB. Single threaded random pointer chasing does not reach the full L3 before latency climbs, because the random 64 byte stride thrashes the translation lookaside buffer, whose reach with 4 KB pages is limited, and it stresses L3 associativity and replacement. Effective capacity under this access pattern is roughly half the nominal size. This is expected behavior [2], and the sysfs cross check is designed to flag it rather than smooth it away. Large pages would isolate a purer DRAM latency, at the cost of measuring something less representative of ordinary code.

6.2 The bandwidth wall

Bandwidth saturates at four to five threads. This is the defining feature of a memory bound workload on a dual channel desktop: the DRAM channels are the bottleneck, and once they are full, more cores cannot help. The practical consequence is that the P versus E core question is moot for STREAM, since the knee arrives long before the core count does. It would matter for a compute bound workload, where the E cores add throughput, which is visible in the all core FLOPS number.

6.3 Why the tile prediction misses

The BLIS analytical model assumes a packed micro kernel in which the L1 resident panel of B is only N_r elements wide. The validation kernel here is deliberately naive: it does not pack, and its inner loop spans the full width of B . Each reused row of B is therefore the full matrix width, several kilobytes, so only a handful of rows fit in L1 or L2 at once, and the optimum moves to a much smaller K_c than the packed model wants. The prediction is not wrong for the kernel it describes; it is describing a different kernel. Closing the gap means implementing the packed micro kernel the model assumes, which is the natural next step and is recorded as future work.

6.4 Limitations

The clock is measured in software, not read from a counter, so it carries a few percent of uncertainty; this is why efficiencies are reported as approximate and a match to within one percent is treated as confirmation rather than a precise claim. Bandwidth uses float arrays and the classical STREAM byte convention. The hybrid P and E identification is heuristic under WSL2. None of these affect the qualitative conclusions, and each is stated where it applies.

Chapter 7

Conclusion

Every headline number a specification sheet quotes for this class of processor was measured directly and cross checked. The cache latencies recover the L1, L2, L3, and DRAM plateaus and agree with the Raptor Cove figures in cycles once the clock is measured honestly. Sustainable bandwidth reaches 87 percent of the theoretical dual channel peak and saturates at four to five threads, exposing the bandwidth wall. Peak single precision throughput matches the ceiling computed from the same measured clock to within one percent. The ceilings form a roofline on which the suite's own kernels sit exactly where the model predicts, and a BLIS style tile prediction, tested against a blocked GEMM sweep, misses the naive kernel in a way that is fully explained by the absence of packing.

The recurring lesson is that the platform, not the arithmetic, is where the honesty is won or lost. On WSL2 the reported clock ignores turbo and the hybrid topology is hidden, and each of these, left unexamined, would have produced a confidently wrong table. Measuring the clock in software and flagging the heuristic topology turned both into stated, understood conditions. The natural continuation is a packed GEMM micro kernel that would close the roofline gap and let the tile prediction be tested against a kernel built on the same assumptions.

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